

# Adapting Data Management for the Digitalization Era

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## Abstract

In an increasing digital world, information systems deal with complex and diverse applications with many associated challenges for modeling, accessing, and manipulating data. We have progressed from traditional data management, to the era of big data management, and now to a digitalization era where the digitalizing of many tasks and processes is having a profound impact on business and society. Proper management of data is, obviously, critical in the digitalization era. However, many traditional challenges of data management remain. This research note proposes a *4S Data Framework* to identify and analyze data issues as they have progressed through the eras. The framework, which is comprised of data semantics, structure, syntax, and situation, is applied to a heavily-dependent, emerging technology, blockchain. The results of doing so help to identify challenges that require a deep understanding of the use of data in the digitalization era, while not ignoring results from traditional data management research.

**Keywords:** Traditional data management, digitalization, big data, 4S Data Framework, blockchain, open environment, closed environment, Change Model of Organizations.

## 1 Introduction

Society functions in an ever-expanding digital world, with many human activities mediated or shaped by digital information. We now live in an era of digitalization, with most functions of business and society critically dependent on data of many different forms and their ability to work together as a whole. As the digital revolution progresses, its impact will be felt in “new and unexpected ways” (Hodson, 2018). This

digitalization of human activities generates large amounts of data, having the potential to be both disruptive and transformative (Larsen 2016).

The success of digitalization is based on the ability to manage data, whether it be traditional or big, newly generated or repurposed, structured or unstructured, or numeric or text. There are many challenges associated with managing data of all types. Besides the sheer volume of big data, for example, there is a need to discover and interpret patterns. Traditional data management has always been concerned with how to capture an accurate representation of business operations. The surge of social media and the need for its interpretation and use for customer management has led to much research on sentiment, and other types of, analysis involving user-generated online content (e.g., Abbasi et al. 2018).

As technology becomes more sophisticated and pervasive, managing data remains critical. Over the last three decades, much research has shown that it is not possible to simply store data randomly. The applications for which specific data sets will be used need to be understood and the data and processing requirements clearly defined. As with traditional and big data management, a deep understanding of the use of data in our increasingly digital world is required. It is, thus, reasonable to examine what is already known in data management and analyze how it relates to the digitalization era. Doing so will make it possible to understand what additional data challenges arise and how they might be addressed. The research question then becomes:

*What data challenges emerge in the digitalization era and how can they be identified and understood from the perspective of traditional and big data management?*

The objectives of this research are to: identify the challenges or issues associated with managing data in the digitalization era; show how some of these can be handled as for traditional and big data; and identify what additional research is needed. The paper first presents important challenges of traditional and big data management, which are presented in a 4S Data Framework of structure, syntax, situation, and semantics of data. It relates these challenges to digitalization using blockchain as an example of emerging technology in this era. The contributions are to: show that traditional and big data challenges remain and continue to impact new technologies; and propose research directions for how to resolve the data challenges.

Section 2 reviews related research on digitalization, organizational change management, and open versus closed systems, needed for understanding the impact of digitalization. Section 3 identifies challenges of data management through a 4S Data Framework. Section 4 applies the framework to blockchain technology to identify existing and new data challenges. Section 5 proposes general data issues in the digitalization era and proposes areas for continued research. Section 6 concludes the paper.

## **2 Related Research**

The digitalization era is driven by advances in technology, data management, and talented researchers, scientists, and practitioners who strive to understand how technology can better our lives. The wider acceptance of, and ability to use, technology also contributes to the pace of digitalization. For business, the focus is on helping companies remain competitive and survive in an interconnected, global, environment.

### **2.1 Digitalization**

Digitization originally referred to the conversion of analog to “digital information and processes” (Loebbecke and Picot 2015). A now much broader concept is *digitalization* that recognizes the transformative role of technology as it digitizes many different aspects of processes and operations across business and society (e.g., Larson 2016). The increasing pace of digitalization means that, as a society, we are changing the way things are done, to which we must all adapt (Baiyere et al. 2017; Bharadwaj et al. 2013; Rodriguez and Piccoli 2018; Weill & Woerner 2015).

According to Gartner<sup>1</sup>, “Digitalization is the use of digital technologies to change a business model and provide new revenue and value-producing opportunities; it is the process of moving to a digital business.” Digital business<sup>2</sup> is “the creation of new business designs by blurring the digital and physical worlds. It promises to usher in an unprecedented convergence of people, business and things that disrupts existing business models -- even those born of the Internet and e-business eras.”

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<sup>1</sup> Gartner IT Glossary.com (<https://blogs.gartner.com/it-glossary/digitalization/>).

<sup>2</sup> FORBES May 7, 2014.

Digitalization is the basis from which fast-paced innovation and transformation will most likely continue for a long time as people and physical devices continue to be connected and used, thus merging digital and physical resources and providing opportunities not available in the past (Tripathi et al. 2019). Baskerville (2019) asserts that “going digital” involves using digital technologies that change “the way things get done,” requiring more creativity on the part of the designer than ever before. Fundamentally, though, digitalization requires the management of data.

The traditional model of information systems is representational, in the sense that the data in a system represents reality in a closed system. Any information system should represent reality by tracking it (Wand and Weber 1993). Digitalization may go beyond this notion and, instead of simply representing reality, creating reality. In artificial intelligence, for example, the intelligence is not human intelligence, but rather a created reality.

## 2.2 Organizational Impact

The notion that digitalization and its underlying technologies have an unending impact on business and society is not new. Leavitt’s (1965) socio-technical Change Model of an Organization, shown in Figure 1, is an early recognition of the interplay amongst an organizational structure, task, people, and technology. In Leavitt’s (1965) model, technology broadly refers to direct problem-solving inventions such as computers. Task is the production of goods and services. Structure is systems of communication, systems of authority, and systems of workflow. People execute organizational work. Leavitt (1965) proposes that, when one of these components changes, there is an impact on the others, which must be realigned to accommodate the changes. The impacts are captured by the arrows in Figure 1.

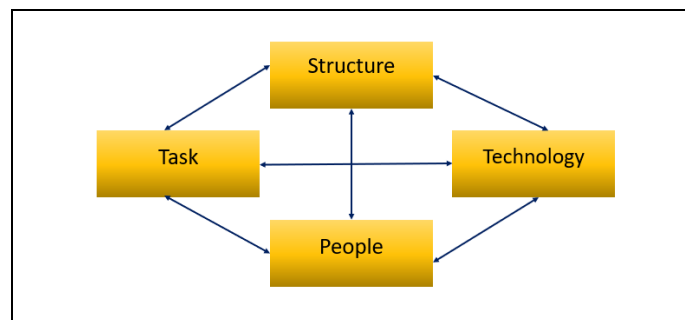


Figure 1: Adapted from Leavitt’s (1965) Change Model of an Organization

The digitalization era has resulted in the technology component having a profound impact on the other components. Leavitt's (1965) model can also be applied to capture how things change. If many processes and procedures are changed by being digitalized, then the challenge is to understand these changes and their organizational impact. Fundamentally, this requires information about the structure, technology, people, and task.

### **2.3 Closed versus Open Environments**

Parsons and Wand (2014) distinguish closed versus open environments. *Closed information environments* are settings in which traditional systems have been managed within well-defined organizational structures and used for the purposes of the organization. Data is “generated, collected, and used almost exclusively within organizational boundaries with specific and well-defined uses in mind” (Parsons and Wand 2014 p.1). If data is missing, a user does not have access to it. *Open information environments* emerge when “users have access to sources over which they have no control.” (Parsons and Wand 2015, p.2). Furthermore, the new sources and uses of data may change rapidly. If data cannot be found in an open system, that does not mean it cannot be obtained. Rather, understanding the context or semantics of data becomes even more important and complicated. Big data has contributed to an open environment with data being more ubiquitous than in prior eras. This will continue in the digitalization era.

Distinguishing characteristics of an open environment are: information diversity (data from multiple, diverse sources); information flexibility (accommodation of data from different, possibly new, sources); and user independence (information generated and used with minimum constraints and control over structure and content). The two main challenges are semantic diversity and data quality (Parsons and Wand 2014). Semantic diversity refers to different users interpreting data or information that is shared in different ways. The structure or sources of the data might even be unknown. Therefore, users need to create different conceptual models and views that can accommodate their own interests and interpretations of the data. Data quality relates to how accurate, complete, and timely the data is (Wang and Strong 1996).

### 3 Data Management

Data management has progressed from developing tools and techniques for traditional, structured data, to managing big data and creating opportunities for novel uses of data from new and emerging technologies.

#### 3.1 Traditional Data Management

Traditional data management focuses on effectively capturing, representing, storing, and using data. Topics include how to create user views, integrate data, manage heterogeneous data, and process mainly structured data. These topics have been studied for several decades. Conceptual modeling<sup>3</sup> was introduced to manage the complexity of traditional data management by focusing on how to abstract and represent concepts from the real world to identify what data to include in a physical implementation (Embley and Thalheim 2012). Although abstractions help to reduce data management challenges, much research has been on ensuring that the data is captured, represented, and stored correctly (syntax and structure), and that the data quality is high and reflects the real-world situation (data semantics and situation or context). The challenges are summarized in the 4S Data Framework in Figure 2 and described below.

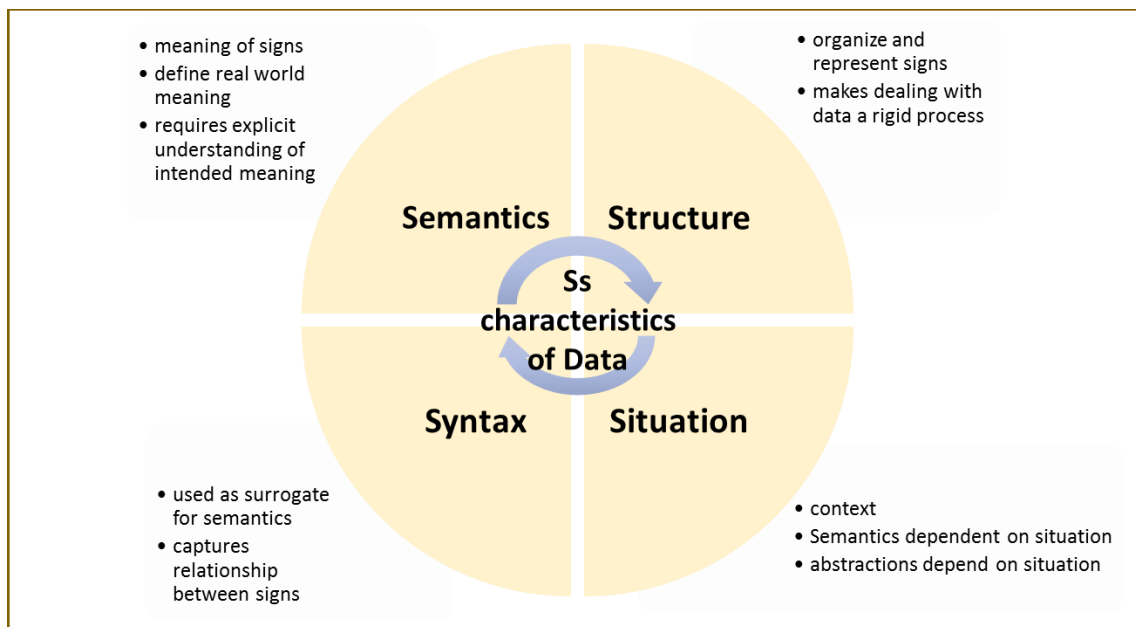


Figure 2: 4S Data Framework

<sup>3</sup> The output of conceptual modeling is a conceptual model, which is usually in the form of a diagram.

The 4S Data Framework is consistent with Semiotics, the theory of signs, which has previously been used to study information systems (e.g., Barron et al. 1999; Stamper et al. 2000). Semiotics is concerned with the properties of things in their capacity as signs (Stamper 1996), and is divided into syntactics, semantics, and pragmatics, which deal with the structure, meaning, and usage of signs, respectively. These corresponds to structure, semantics, and situation in the 4S Data Framework. The “signs” are representation for storage and processing by computers and interpretations by human (Barron et al. 1999), corresponding to the syntax dimension of the 4S Data Framework.

### **3.1.1 Syntax**

Syntax is the set of rules, principles, and triggers that govern the arrangement of data in a database for the purpose of storing and retrieving data. Machines, no matter which kind, operate at the syntax level where there is no meaning. Syntax captures relationships between data and is often used as a surrogate for semantics<sup>4</sup>. It is, generally, not possible to describe a language’s meaning by a computer program. Rather, a machine’s representation, or “abstract syntax” is used to do so (Harel and Rumpe 2004)<sup>5</sup>. A challenge is to identify and use reasonable labels and categories to assign to concepts (Mendling et al. 2010). In open information environments, markup languages create annotations so that concepts can be matched to higher-level or generic concepts; for example, to connect the name of a person to the concept of a person or the notion of being associated with a birthplace.

### **3.1.2 Structure**

Humans naturally organize and represent concepts and signs. Models, in general, are used to capture and represent concepts to support our mental ideas of how a world operates. Conceptual models capture some phenomena of the real world for a specific domain (Murphy 2002) and are governed by their purpose

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<sup>4</sup> In linguistics, syntax is the set of rules, principles, and processes that govern the compositions of sentences in a given language, specifically word order. The term *syntax* is also used to refer to the study of such principles and processes, with the goal of many syntacticians being to discover the syntactic rules common to all languages.

<sup>5</sup> A language consists of “a syntactic notation (syntax), which is a possibly infinite set of legal elements, together with the meaning of these elements, which is expressed by relating the syntax to a semantic domain.” (Harel and Rumpe 2004, p.65).

(Thalheim 2012). Although models provide structure, they are, in essence, knowledge representations captured by syntax. Data models and data structures are intended to make dealing with data a rigid process.

### **3.1.3 Semantics**

Semantics refers to the meanings of words and phrases within a particular context (Merriam-Webster.com). Semantics define the real-world meaning, or essence, of the terms used in a model; that is, of the words and phrases selected to label a concept intended to capture a real-world phenomenon (Storey 2005). Some semantics might be well-known; others are dependent upon a given situation and can have more than one interpretation (Mendling et al. 2010). Semantics, as the study of meaning, deals with the relationship between signs and the things in the real world to which they are applicable. Therefore, one must know to what a sign can be used to refer to, in order to grasp its meaning.

Conceptual modeling builds a “representation of selected semantics” of a domain (Weber 2003, p.1), although modeling semantics remains difficult task. Of the various approaches to dealing with semantics, an important challenge is how to identify the surrogates (e.g., syntax, data abstractions) that can be employed (Barron et al. 1999). Any attempt to represent semantics must involve the object being represented, the representation mechanism, and a means for interpretation (Abdoullaev 2008).

Early research on data abstractions attempted to address the problem of semantics by developing rules and guidelines for using data abstractions in the database design process (e.g., Teorey et al. 1986). Most of these rules, however, identify relationship and cardinalities, and then mechanically apply them to translate a conceptual design into a logical design for implementation. Semantics are often dealt with by using “built-in” semantics, as in data abstractions, or by mapping to some external source, such as an ontology or lexicon for interpretation (Guarino 1998).

### **3.1.4 Situation**

A useful information system that deals with data, whether big or small, structured or unstructured, needs knowledge of how things behave and how people comprehend them (Cambria and White 2014), making the situation or context important. Knowing the situation can help to reduce the complexity of capturing semantics because certain semantics can be implied from a known situation. Similarly, what is being



abstracted in a conceptual model can be simplified if the situation is known. Consider the well-known *is-a* relationship. Suppose *Presenter is-an Attendee* occurs in a conceptual model. There might be a missing level(s) of abstraction between attendee and presenter, such as the mutually exclusive subclasses of professor and student. Such modeling decisions require situation knowledge of how the real world operates. The challenge is that some semantics are well-known, but others vary depending upon the situation (Mendling et al. 2010).

### 3.2 Big Data Management

Business intelligence and data analytics initiatives are found in all industries (Chen et al. 2012, Dahr and Agarwal 2013, Goes 2014; Gupta and George 2016). At the heart of this activity is data and its proper and effective management.

The big data phenomenon has generally been described by the three Vs of volume, velocity, and variety. The two best-known “additional Vs” are veracity (accuracy) and value. Challenges arise in that the volume (scale) may be so large that it is difficult for human decision makers to understand it, and requires advances in processing capabilities. The velocity is practically too fast (Embley and Liddle 2013). Perhaps, most importantly, the value is difficult to ascertain (Storey and Song 2017). Modeling and representing big data are not straightforward activities for specific applications. To realize real-time analysis and decision making, well-known goals when employing big data, complete interoperability among data sources must be established so systems can interact seamlessly with each other. This involves challenges of semantic integration of interrelated information from multiple sources to obtain a shared understanding (Obrst et al. 2014).

The big data era has attempted to identify, extract and interpret *patterns* in data. With the recognition of the potential value of text data, processing both structured and unstructured data are important, as are advances in data mining, machine learning, sentiment analysis, and other efforts (Maass et al. 2018). Social media data, especially, is being analyzed because of its potential use to gage a customer’s response to a company, or to mine a competitor’s situation. Much social media data, however, is unstructured, and its context often unknown or cannot be inferred from the data alone.

Many challenges of big data are the same as traditional data management. They do not go away simply because the data is ‘big.’ These challenges remain unsolved, despite advances in the big data era. For example, the data semantics problem, which is unresolved in traditional data management for structured data, is now extended to unstructured data. Thus, the data challenges of big data, as captured by the “Vs”, can be extended to include the traditional characteristics, as characterized by the 4 “Ss” shown in Figure 6.

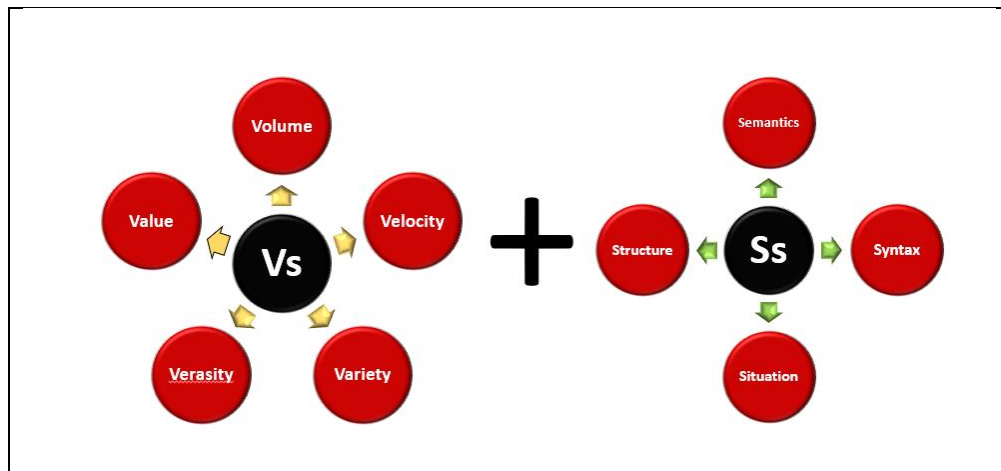


Figure 6: Adding Ss to the Vs of Big Data

### 3.3 Digitalization Data Management

The digitalization era cannot ignore the characteristics of big data in open environments and the data challenges of traditional data management in closed environments because they are part of our ecosystem. Digitalization impacts entire organizations, so the resulting organizational changes must be understood.

The multi-dimensional nature of digitalization data management is shown in Figure 7. The 4 Ss dimension came from traditional data management and remain in the big data and digitalization eras. The same applies for the social-technical dimension, which incorporates the components of organizational impact. This dimension also represents the fact that, progressing from syntax to situation is at an increasing level of difficulty and that, as we progress from people to technology, there are more opportunities for automation. Finally, the closed versus open environment dimensions reflects the availability of data, with open environments being significantly more complex than closed environments.

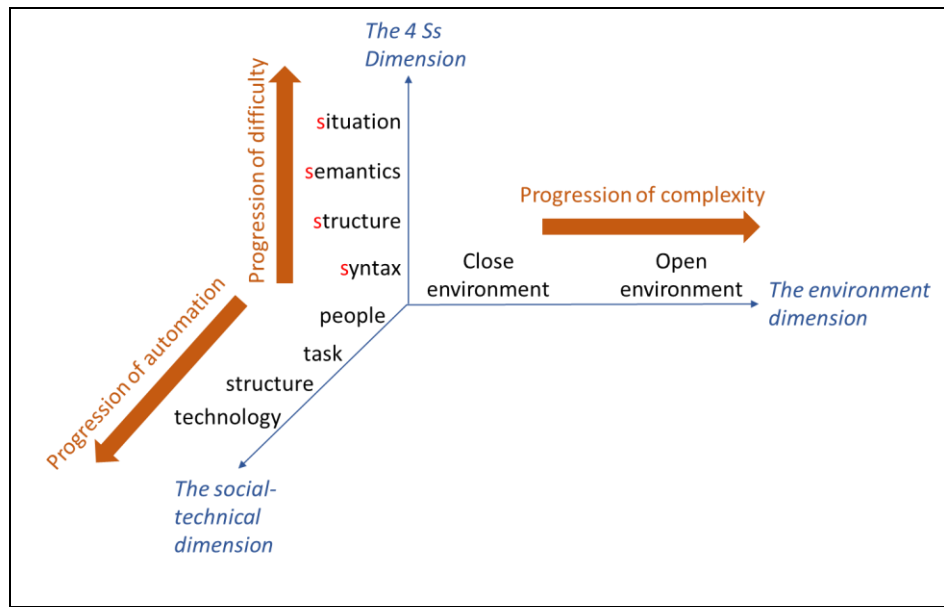


Figure 7: Digitalization Data Management

The dimensions are discussed further below with respect to an emerging technology. The challenges that arise are also a research opportunity to provide guidance to progress the digitalization era, taking advantage of what is known about traditional and big data management and the characteristics of newer data technology.

#### 4 Blockchain as Emerging Technology in Digitalization Era

Blockchain, as an emerging technology in the digitalization era, is data intensive. Forbes regards it as going hand-in-hand with digital transformation<sup>6</sup>. Blockchain is highly regarded as having the potential to transform information-based operations. It is a distributed network of peer to peer, encrypted public and private ledgers, composed of data records in “blocks” that are linked into an immutable chain that is verified and managed using smart contracts to execute transactions automatically. Blockchain’s architecture is intended to support immutable trustworthy records without needing a trusted third party (Beck et al. 2017). Because of its perceived trust, blockchain is considered well-suited for applications such as financial services, healthcare, and government services (Tapscott and Tapscott 2016; Lemieux et al. 2019). The result

<sup>6</sup><https://www.forbes.com/sites/danielnewman/2017/10/24/blockchain-and-digital-transformation-go-hand-in-hand/#843de0f46f7e>

is enthusiasm for blockchain by practitioners, government organizations, and academics (Glaser 2017), which is often regarded as potentially disruptive.

Still, blockchain is just a technology with many of the same implementation issues and challenges as any other technology, including how to manage effectively the large amounts of data it generates and processes. Since the data can be generated in a short period of time, blockchain displays many of the characteristics and challenges of big data (Garimella and Fingar 2018), even though Blockchain applications may have more structured data.

In the digitalization era, data needs to be understood from both the point of view of storing and using it, similar to traditional data management. However, data cannot be stored randomly. It is important to identify what data will be used, how it should be stored, and what obstacles might be faced when using it. Furthermore, data is increasingly ubiquitous. This does not necessarily mean more centralized or more decentralized. It does, however, imply the need to manage more data, which might require work that falls somewhere between traditional data management and big data management. For example, instead of waiting for the data needed for an application to become accessible in an open environment in a format that lacks structure, semantics, or situation, it might be feasible to capture these characteristics in the data, and make sure that data is not stored randomly.

#### **4.1 The 4 Ss Dimension**

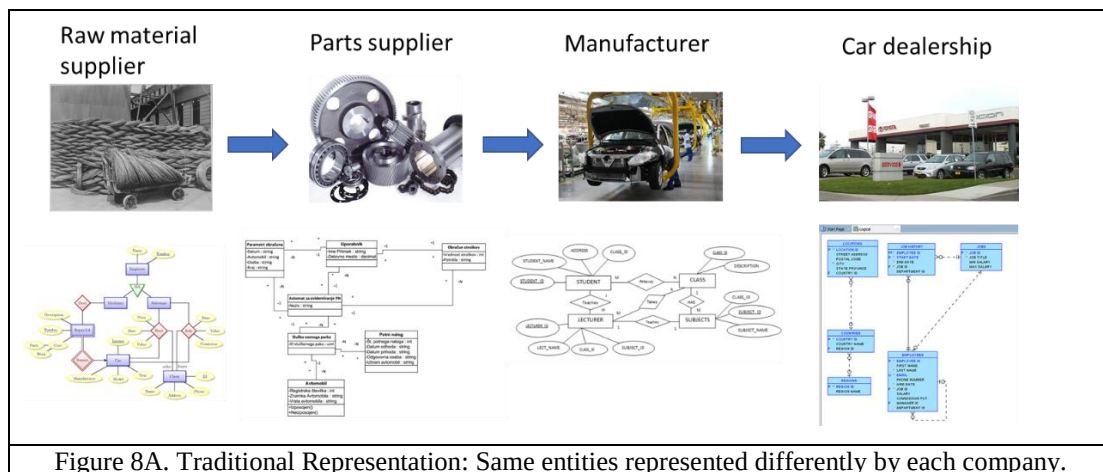
In blockchain applications, data is captured from transactions recorded in a distributed ledger. These transactions have a mixture of traditional and big data, especially those from permissionless applications, because anyone can participate in them. Blockchain is analyzed from the challenges of semantics; structure (via Leavitt's model); syntax; and situation (closed versus open environment).

##### **4.1.1 Semantics**

Semantics must be addressed differently in blockchain applications than in traditional information systems. In traditional, centralized, database systems, it is difficult to reconcile semantic interpretations of the data, especially to deal with semantic interoperability (Lemieux, 2017). On the other hand, every member of a

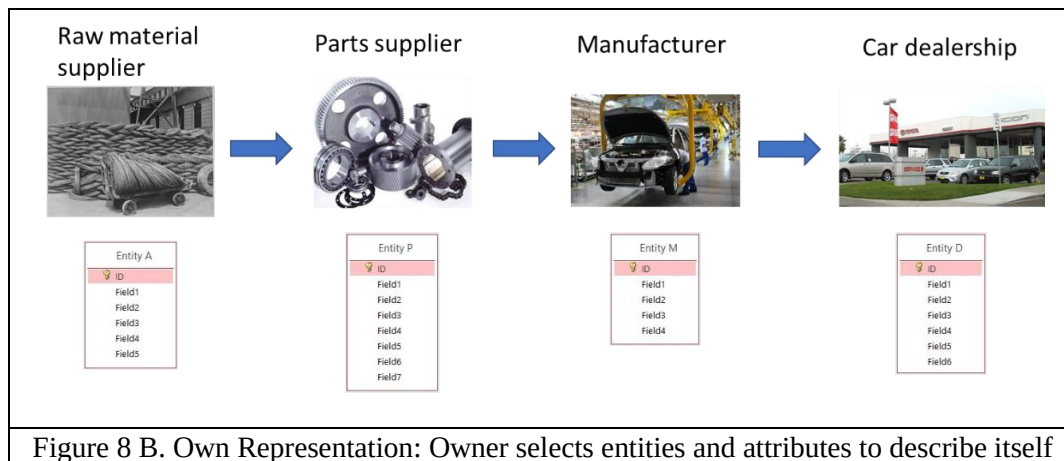
blockchain has the same copy of a transaction, so this, in some sense, “solves” the semantics problem. There are, of course, protocols that must be in place to enable such behavior.

A major problem of traditional information systems is that different systems represent the same data differently (and represent different data using the same term). This problem occurs because the same physical thing or concept is viewed differently by different people for their own usage and situation. If systems are not required to interoperate, this silo representation is acceptable. In the digitalization era, however, a silo representation magnifies the interoperability issue. Given that the data representation of blockchain is in its infancy, a new approach is needed. For example, blockchain is decentralized and requires multiple organizations to work together. One way to accomplish this is for individuals to represent their own information in their own way, which dictates how they want others to view their information. When others need the information, they can then request it from the information owner. This difference is illustrated using a car supply-chain. Figure 8A shows the traditional representation.



Here, the supply-chain is simplified to two upstream suppliers for a car manufacturer (parts suppliers, who obtain their input from raw materials suppliers) and one downstream customer, which is the car dealership. Traditionally, the car manufacturer would have a database representing information about its suppliers and dealerships. A parts supplier would create its own representation for its raw materials suppliers, the car manufacturers, etc. Each of the players in this supply-chain is being represented multiple times using different representations by others, making interoperability challenging.

Now, consider the situation where the car manufacturers represent only themselves and do not define how they see suppliers. Instead, suppliers tell car manufacturers how they should be represented. If car manufacturers need information about parts, suppliers or dealerships, they request them from the corresponding stakeholders. In this way, the semantics is defined by the owner of the entity, not by how others perceive that entity. This greatly simplifies the data semantics issues, by taking advantage of the decentralized and distributed nature of blockchain. This is shown in Figure 8B.



To facilitate the owner of the entity representing itself, the best way to implement the entity is by *object-oriented technology* and perhaps supplemented with *semantic web technology*. Object-oriented technology facilitates incremental and decentralized implementation of entities; semantic web technology provides mechanisms to interoperate in an open system environment (e.g., through tagging).

As more and more data are managed through blockchain applications, there is a need to reconsider some basic data management concepts. For example, if an entity, such as a product item in a supply-chain, exists within a blockchain, then a primary key of the product item might no longer be needed<sup>7</sup>. The lack of the need for a designer-specified primary key is a major departure from traditional data management.

The semantics of transactions are usually not documented, but, rather, somewhat buried in the application program, so no one outside the organization will be required to know or understand these

<sup>7</sup> This is because the entity's identity can be identified in the blockchain as represented by the sequence of transactions, through which the entity passes.

semantics. This is a consequence of information systems operating in a closed world. However, blockchain operates in an open world, where the semantics of a transaction must be transparent; otherwise, no one else will know what has been done. Even if the domain is known (e.g., supply chain of car parts) and provides enough domain semantics, the transaction should also provide the context (situation) necessary to achieve this transparency.

Every transaction inherently has semantics. These can be made explicit by identifying a transaction's social-technical characteristics using Leavitt's (1965) Change Model of Organizations. The *transaction* is the task. The *structure* is the part of the workflow this transaction executes. The *people* are responsible for the success or failure of this transaction, so the identification of the person must be recorded. Blockchain, of course, is the *technology*. The combination of these four social-technical components, in essence, serve as a surrogate for semantics and, in this way, operationalize the semantics of a transaction. Furthermore, one way to appreciate the organizational impact is to specify explicitly these characteristics for each transaction (i.e., the structure, actor, task to which the transaction is being applied, and technology). Since Leavitt's model (1965) deals with change, and is well-accepted, specifying these four aspects, should support future change and make the blockchain application more agile.

#### **4.1.2 Structure**

The structure of blockchain is often described as a "decentralized system," meaning that the processing of blockchain transactions occurs over multiple nodes. These nodes can be combined to form a distributed database. This decentralized structure has implications for the fundamental operations of blockchain. Chaining is the manner in which sequencing is carried out in a blockchain transaction. Hashing and chaining, together, provide sequencing which give the blockchain its structure. Hashing involves taking an entire, previous block, and reducing it to a few bits which become the hash for the next block. This is very similar to how digital signatures work when an original document is reduced to, perhaps, 15 characters, becoming a surrogate for a person's signature that is attached to the document.

There is no structure to the transaction other than the database used to store it. This leaves some flexibility to impose a structure onto the transactions. One node is selected to process a transaction, where

the (final) result of the processing is replicated in all the other nodes, and stored in a database, thus showing the relationship between the transaction and the database.

A well-known characteristic of blockchain is a “smart contract.” A smart contract has a set of conditions, which, when met, allows the transaction to be executed. It is somewhat analogous to the notion of constraints and triggers in traditional data management, which represent business rules. A smart contract is simply a program and a syntactical way to try to capture some of the semantics of the real world. If the number of smart contracts becomes large, and some of them are inter-related, then some organization of the smart contracts is needed to manage the volume. One way to manage the volume is to use, for example, is-a (in UML, it is <<extend>>) and part-of (in UML, it is <<include>>), which are part of the structure of smart contracts.

In a student example, there must be a transaction in blockchain to show that someone is a student and whether that person is a teaching assistant. Blockchain has self-sovereign identities. If a student is a teaching assistant, then there will be multiple transactions related to being a teaching assistant, such as grading assignments, meeting students, and so forth. Understanding how these transactions are structured and can be used to infer a person is a teaching assistant may require more research on instance-based data representation<sup>8</sup> (Parsons and Wand 2000) and its relationship to smart contracts or transactions.

### 4.1.3 Syntax

The syntax in blockchain forms the basis for the rules that govern the composition of transactions, how they are stored in the (distributed) database, and how they can be retrieved. At a high level of abstraction, the syntax is not complicated. The way the data are stored and retrieved in the distributed database is part of a smart contract. A smart contract is constructed by identifying the pre-defined considerations such as the variables and conditions. Then, the events (transaction, retrieval) that trigger a contract execution are identified. The terms of the contract dictate movement of value based on the conditions that are met.

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<sup>8</sup> This is representing an instance independent of the class or classes they belong to. The instance-based approach facilitates dynamic or temporary association to classes, and can be retrieved without belonging to a class.



Changes to the records are made in a settlement (for records that might be either on or off the chain). Since a great deal of research has been carried out on constraints and triggers in traditional data management, exploring and adapting appropriate ones for smart contracts can, thus, advance blockchain technology based upon previous data management foundations.

#### **4.1.4 Situation**

Some transactions in blockchain can easily be carried out without concern for the situation. For example, in bitcoin, a command which indicates that one person wants to transfer money to another, is independent of the situation. In a land title transfer, all that is required is to know the buyer and the seller. In these situations, there is little concern for either situation or semantics. It is simply an efficient syntax operation.

An example where the situation (context) is important is medicine traceability. There are initiatives that attempt to use blockchain to trace medicine, all the way from the initial research on the medicine to the people using it. The results could help to understand known side effects in order to assess the effectiveness of a certain medication in that medicine's life cycle. Blockchain does this quite effectively. In a medical application, the patient owns his or her data and, therefore, has some inherent notions of the semantics attached to it. Note that this is a reversal of roles for the patient and establishment that is treating the patient.

If contextual information can be identified, then, it can be added to whatever mechanism is used to represent semantics. In an open environment, data may arrive at any time. The data people provide might have different context, etc. Blockchain is somewhere in between because it needs to be part of a chain, thus, following the context of that chain. It is dependent upon the application domain, but, at the same time, can be permissionless, a characteristic of an open environment.

Understanding the situation or context is important for any information systems development effort. The examples in Table 1 illustrate how blockchain is situation dependent.

Table 1: Blockchain Characteristics Applied to Multiple Domains

| <b>Blockchain<br/>Concept/characteristics</b> | <b>Application</b>                                             |                                                      |                                                                      |
|-----------------------------------------------|----------------------------------------------------------------|------------------------------------------------------|----------------------------------------------------------------------|
|                                               | <b>University</b>                                              | <b>Healthcare</b>                                    | <b>Financial</b>                                                     |
| Transaction                                   | Student received grade for course (grade stored in blockchain) | patient gets lab test (results stored in blockchain) | Customer transfers money (amounts and accounts stored in blockchain) |
| Participant                                   | Student, professors                                            | Patient, physician, lab technician                   | Customer, lender, financial institution                              |
| Smart contract                                | When requirements fulfilled, certificate issued                | Test results sent to appropriate physicians          | Credit alert sent to appropriate manager                             |

A closed environment exists when all data is generated and maintained within an organization. In an open environment, anyone can access the data (e.g., using the Internet). Blockchain has characteristics of both. It is situation-dependent, analogous to conceptual modeling. For example, a student who wishes to attend another university would need to convert their course credit to that of their home school. In an open environment, a student can attend any, and/or multiple, universities and, once that student has obtained the total required credits, he or she can obtain a degree.

In fact, blockchain has been analyzed for this very application (Garimella and Fingar 2018). Every time a student takes a course or completes an assignment, there is an entry in the corresponding blockchain. (Everything is noted in the blockchain.) Provided the university a student attends (or take a course from), is within the blockchain, it is perfectly acceptable for the student to complete his or her degree requirements in this manner. Such an arrangement would be extremely useful for part-time students.

Part of the challenges of semantics may be due to the nature of open environments. The number of domain areas (e.g., healthcare), knowledge areas (e.g., archival science), and technology areas (e.g., computer science) will present a level of complexity, previously unseen. Preparing and using data will require non-traditional collaboration. For example, blockchain inherits many properties of both closed and open environments. Addressing the inherent challenges of an open environment will require knowledge

from many different, but relevant, disciplines such as linguistics (for semantics), ontology (for mappings), organizational science (situation/context), and computer science (databases).

Table 2 summarizes the 4 Ss Data Framework as it relates to blockchain.

Table 2. Blockchain's Data Characteristics

| <b>Data Characteristics</b> | <b>Blockchain</b>                                                                                                                                                                                                                                                                  | <b>Relevant Existing Research</b>                                                                                                                                                                    |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Syntax                      | Relate syntax to a smart contract. No change. Just another technology with its own syntax.                                                                                                                                                                                         | Database constraints and triggers                                                                                                                                                                    |
| Structure                   | Data is structured. Captured from distinct transactions.                                                                                                                                                                                                                           | The instance-based data representation (Parsons and Wand 2000; Lukyanenko et al. 2015).                                                                                                              |
| Semantics                   | No longer necessary to create a representation of a concept and then use it. Semantics captured better because people (for example) present themselves and do not rely on others to do so for them (e.g., patients understanding of semantics should be same as that of hospital). | Object-oriented technology and semantic web technology                                                                                                                                               |
| Situation                   | Capturing the context requires less representation because one can always goes back to the owner (e.g., in office visit, context important to patient, such as purpose of visit).                                                                                                  | Knowledge from many different, but relevant, disciplines such as linguistics (for semantics), ontology (for mappings), organizational science (situation/context), and computer science (databases). |

#### 4.2. The Environmental Dimension

Interoperability in open environments is needed. Traditional, existing, databases will not disappear. They will still be used; the same holds true for big data such as those in social media. In the digitalization era, they need to be interoperable in order to avoid replacing many of the existing information systems, which are expensive to build and maintain. Legacy systems will not be terminated simply because of new, open environments. In a traditional database system, interoperability problems are a result of databases being developed independently (in silos). Since blockchain promotes peer-to-peer interactions, and for this to be possible, interoperability needs to be considered ahead of time, so it can be represented or captured appropriately (Lemieux et al. 2019).

Blockchain is semi-open and can alleviate the interoperability issue if performed properly. With digitalization, data will become even more ubiquitous (not necessarily more centralized, or more

distributed). To digitalize certain activities, structured (or semi-structured) data is required. Knowledge from traditional heterogeneous database integration could be useful. Since blockchain is a relatively new, but emerging, technology, it is necessary to study how existing knowhow (e.g., database integration) might be utilized and the challenges of doing so.

#### **4.3. The Social-technical Dimension**

The technology component of Leavitt's (1965) Model of Organizational Change has now become dominant when considering the impact of the other components. The people are the users of blockchain technology. Certain tasks, such as those that require a high degree of trust and a high volume of transactions are ideally suited as an application of blockchain technology, such as those in healthcare applications. The structure of the organization is now supported by distributed technology. More importantly, an organization becomes dependent on other organizations over which it might have no control. Thus, a balance of the social aspects of people and society as well as technical aspects of organizational structure and processes, is needed.

### **5 Discussion**

Digitalization involves the use of data in the transformation of once-manual activities and processes to computer-supported ones. To understand the impact of digitalization, we analyzed the progression from traditional data management to data management in the digitalization era. Regardless of how this progression emerged, data remains the fundamental input that needs to be managed, making the following essential for continued data research.

**The Digitalization Era Can Benefit from Adapting Existing Data Management Knowledge.** The need for data management does not change throughout the different eras. Rather, data management techniques used in one era can be used in other eras, although they may need to be improved or simplified, but not recreated. Researchers have a great opportunity to apply existing knowledge (e.g., traditional data management and big data analytics) to interesting, and often complex, problems that continue to emerge in today's society. Examples are the adoption of blockchain technology in healthcare or agriculture (Kamilaris et al. 2019) or machine learning applications. It is important not to become lost in the hype new

technologies. Rather, research in data management leads to the continuous accumulation of knowledge. Researchers need to be mindful of new technologies, but not devalue existing data management knowledge.

**The Digitalization Era Requires Attention to Decentralization.** Technologies in the digitalization era are decentralized. Consider four common technologies that are part of the digitalization of business and society: artificial intelligence, augmented reality, blockchain, and Internet of Things<sup>9</sup>. Additionally, social media exhibits characteristics of being complex and highly unstructured. These technologies share a common characteristic of being decentralized because they operate autonomously, but need to cooperate at some point in time during their use. The open system environment concept, used to describe the big data era, is also applicable to the digitalization era, and consistent with this observed decentralization. This conceptualization of decentralization provides a practical way to focus on relevant literature, foundations, and research approaches. For example, the instance-based approach (Parsons and Wand 2000), as suggested for an open systems environment, can potentially be used to represent data in the digitalization era.

**The Digitalization Era Compels More Experimentation to Accumulate Knowledge.** Figure 7 highlight the interplay among the 4Ss, the social-technical (organizational change), and open versus closed environments. This leads to the cone shape in Figure 9, which illustrates the feasible “solution-space” for where advances in digitalization data management can be made. For example, it is nonsensical to only work on syntax (in the 4 Ss) and actor (in Leavitt’s model) when a (data) problem is a complicated one in an open environment (in the open versus closed dimension), because it is not in the cone solution space. This is because open environments require emphasis on semantics and situation (in the 4Ss framework) and on structure and automation (in Leavitt’s organizational model). On the other hand, in a very closed environment, working on syntax and actor alone might be sufficient, as is the case for a brick and mortar business application. The syntax, actor, and closed environment solution space is small and, as shown in Figure 9, represented in a rather narrow tip of the cone. On the other hand, the wide-open end of the cone is too open-ended, too complex, and too difficult, although the solution space is large. Perhaps the most

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<sup>9</sup> <https://www.ledgerinsights.com/blockchain-enterprise-digital-transformation-survey-okta/>

valuable and reachable solution space is in the middle part of the cone, which, in essence, represents data challenges in the digitalization era.

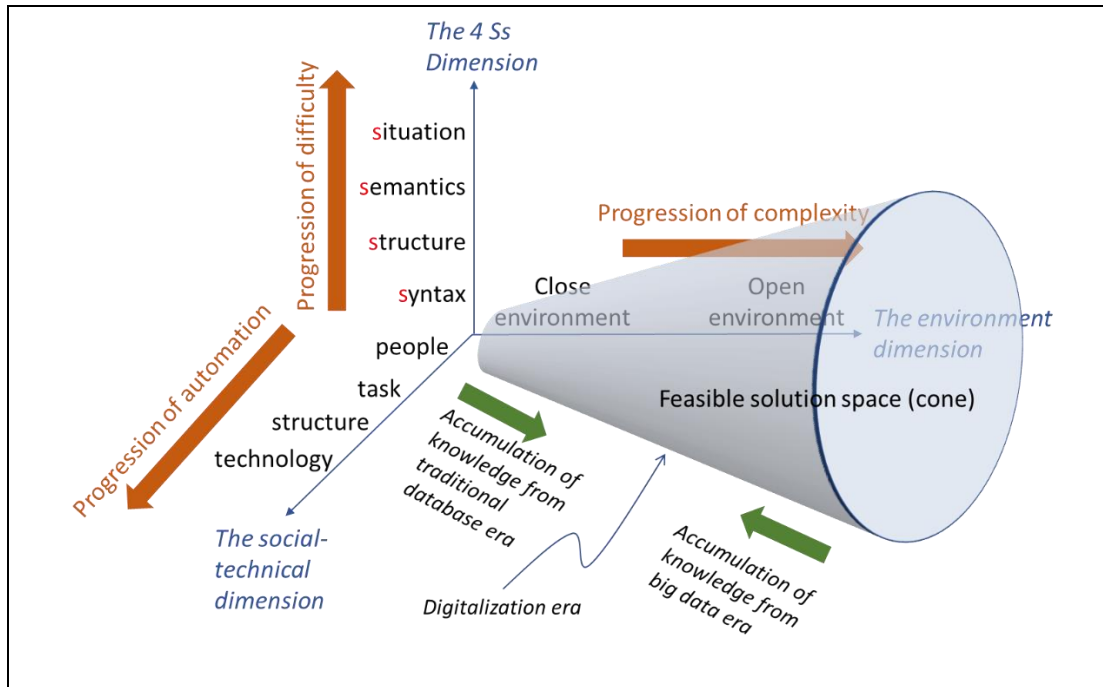


Figure 9: Digitalization data management: Feasible solution space

Digitalization data management is somewhere in between traditional data management and big data eras (i.e., middle of the cone in Figure 9). The structure and semantics of the 4 Ss are the most important, the environment is partly closed and partly open, and the task and structure can be used to mediate people and technology in the social-technical dimension. Note that the 4 Ss magnify the data management challenges in the digitalization era because of semantic diversity and data quality.

The optimal point in the cone for the digitalization era is, of course, unknown and subject to further research. To manage the complexity and difficulties associated with finding this optimal point, resolving data challenges in the digitalization era will require abstractions and assumptions. This will lead to more trial and error and experimentations, in research endeavors. More research opportunities mean that a framework is needed to tie everything together to facilitate the accumulation of knowledge. Figure 9 could serve as one such framework, or the starting point of another.

**The Digitalization Era Necessitates Interdisciplinary Collaboration.** We are rapidly reaching a stage in technology advancement where one discipline cannot progress a field significantly by itself. Information systems (IS) has often borrowed and adapted theories and frameworks from neighboring fields. This has been possible when the data needed to deal with a problem was contained within a closed system. Moving to a complex, open environment, means it is no longer possible to borrow or adapt ideas and theories from a single neighboring field. A healthcare application in blockchain, for example, requires input from physicians, pharmacies, lawyers, patients, insurance companies, and so forth. For complex applications, it is impossible for researchers in information systems to learn everything needed from multiple, relevant fields (psychology, computer science, medicine, etc.). Research in the digitalization era will, instead, require multiple experts working on problems collaboratively, and possibly, across disciplines.

Obvious areas for collaboration where data has been used extensively include archival science, organizational science, and linguistics. In an open environment, data is used in many different ways by many users in different applications. The users' needs should be represented and stored. Archival science is concerned with longevity and how people can retrieve data in the future (e.g., a few hundred years from now) and still be able to understand and use it. Organizational science focuses on the interaction, impact, and interdependence of organizational components. There are many organizational learning models and frameworks, including Leavitt's (1965) model, as illustrated in this paper. A careful selection and utilization of the relevant models and frameworks for data semantics, for example, could be a promising research direction. Linguistics, is concerned with semiotics, and could, thus, provide input from a language point of view. Of course, the area of information systems, itself, focuses on traditional data management and how that work can now be applied to big data and emerging technologies.

**The Digitalization Era Needs Academic Leadership.** The magnitude and impact of digitalization will, no doubt, continue to be felt for a very long time. It is easier to manage digitalization properly the first time, rather than deal with the fallout of improper management. Much effort is needed to create, store, and manage data that will be useful in data analysis in the digitalization era. The basic theme of this article is

how to bridge the gap between traditional data management work and its evolution into the digitalization era. Clearly, information systems research and leadership are needed.

This paper focuses only on data. Obviously, digitalization is not just about data. The continued societal impact needs to be studied further. The organizational impact is large. Hudson (2018) notes that this impact needs to be broadened to society, as a whole, as “the underlying technology becomes more sophisticated and pervasive.” Digitalization causes changes in representation, procedures, and processes. Some of these issues may be easy to address; others may be difficult and complication. Regardless, the impact still requires much research that focuses on both the technical and societal impacts.

## 6 Conclusion

The digitalization era, as a continuation of our increasingly digital society, has been bringing about disruption, as well as the potential for innovation and future advancement. Data is no longer confined to corporate databases, as in traditional, closed environments, but is now available anywhere, in open environments, thanks to the ubiquitous nature of the Internet and its adoption. This research has examined data in the digitalization era to deepen our understanding of its challenges and to demonstrate that research results from traditional data management remain applicable. The unique characteristics of the digitalization era were identified by applying notions of organizational change and open versus close information systems environments. A 4S Data Framework of syntax, situation, semantics, and structure was derived from traditional data management challenges and evolved to include a social-technology and an environment dimension. The result is a cone-shaped representation that can lead the discussion of future research.

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